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The investigation of a μGal-level cold atom gravimeter for field applications

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Abstract

Typical atomic gravimeters have a heavy magnetic shielding to avoid the quadratic Zeeman effect, and also a complicated active vibration isolation system to suppress the vibration noise. Both of them make it difficult to build a mobile and compact gravimeter. In this paper, we present the implementation of an atomic gravimeter aiming at field applications. Our gravimeter uses improved magnetic coils instead of the expensive mu-metal for magnetic shielding. The quadratic Zeeman shift is evaluated with high accuracy. Moreover, a portable platform of relatively small size is applied for vibration isolation. The total interrogation time is optimized to 120 ms and the repetition rate is 2.2 Hz. A sensitivity of $1.0 \times 10^{-7}\ g\ \mathrm{Hz}^{-1/2}$ and a resolution of $5.7 \times 10^{-9}\ g$ within 1000 s integration time are reached. A continuous g measurement over 128 h is carried out. Moreover, a whole seismic wave of about 1 h that occurred in Pakistan on 28 September 2013 is recorded by our atomic gravimeter. The results coincide with that recorded by a traditional seismic detector very well.

Keywords: atomic gravimeter, the quadratic Zeeman effect, atom interferometer

(Some figures may appear in colour only in the online journal)

1. Introduction

The performance of atom interferometers has been improved substantially over the last two decades. Applications of such an interferometer are growing in fundamental physics. For instance, it has been employed in the measurement of the fundamental constants [1–3] and proposed for tests of general relativity [4] and the equivalence principle [5], and even for the search for gravitational waves [6]. Meanwhile, in the field of applied physics, quantum inertial sensors based on an interferometer, such as atomic gravimeters [7–9], atomic gyroscopes [10] and atomic gravity gradiometers [11, 12], have attracted worldwide attention due to their remarkable stability, high sensitivity, high repetition rate and potentially high accuracy. All these advantages make them suitable for practical applications in metrology, geodesy, inertial navigation and even for promising space missions [13, 14].

Much effort has been made to simplify the laboratory-based experimental apparatus so as to build compact and transportable instruments [15, 16]. Recently, mobile atomic gravimeters have become practical, benefiting from the rapid development of the related technologies [17, 18]. The sensitivity and accuracy are comparable to those of the best classical gravimeter (FG5 [19]). However, these mobile gravimeters are relatively heavy and expensive. One reason is the existence of magnetic shielding, which usually consists of several layers of mu-metal. For our atomic gravimeter, improved coils are utilized to compensate the Earth's magnetic field at the centre of the three-dimensional magneto-optical trap (MOT). To define a quantization axis in the interference region, two coils are applied to produce a homogeneous bias field. In order to improve the measurement precision of an atom gravimeter, a vibration isolation system is necessary. However, isolators are usually very complicated due to the active feedback apparatus [17, 20, 21]. In our case, the vibration noise of high frequency is well suppressed by a passive portable platform.

For our atomic gravimeter, with a total interrogation time of 120 ms and a repetition rate of 2.2 Hz, a sensitivity of

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$1.0 \times 10^{-7}\ g\ \mathrm{Hz}^{-1/2}$ has been achieved. The tidal phenomenon is observed by monitoring the local gravity continuously over 128 h. Moreover, a whole seismic wave that occurred recently was recorded in great detail with our atomic gravimeter and the results are compared with that recorded by a traditional seismic detector; they coincide with each other very well.

2. The principle of measurement and experimental apparatus

The principle of an atomic gravimeter has been discussed in great detail [22]. The separation and reflection of the atomic matter waves are realized by inducing two-photon Raman transitions [23]. In our experiment, the Raman transitions couple the two ground hyperfine levels of ^{87}Rb ($5^2S_{1/2}, F = 1, m_F = 0$ and $5^2S_{1/2}, F = 2, m_F = 0$). Two counter-propagating laser beams of wave vectors $\vec{k}_1$ and $\vec{k}_2$ vertically illuminate the cold atomic cloud. A Doppler-sensitive Raman configuration is selected so that a large momentum of about $2\hbar\vec{k}_1$ can be transferred to the atom. The atomic matter waves are split, reflected and recombined by a sequence of $\pi/2$–π–$\pi/2$ Raman pulses. Suppose all the atoms are in the initial state $|F = 1, m_F = 0\rangle$. The first $\pi/2$ pulse drives the atoms into an equal superposition of states $|F = 1, m_F = 0\rangle$ and $|F = 2, m_F = 0\rangle$. Then the atoms fall freely in the gravitational field within time T. A π pulse as a reflector is applied to completely transfer the atoms from one state to the other. After another free flight time T, the second $\pi/2$ pulse is applied to make the two wave packets interfere with each other. At the end of the interferometer, the interference fringes can be observed and the transition probability P can be described as

$$P = (1 \pm \cos(\vec{k}_{\mathrm{eff}} \cdot \vec{g} - \alpha)T^2)/2. \quad (1)$$

Here, $\vec{k}_{\mathrm{eff}} = \vec{k}_1 - \vec{k}_2$ $\left(\left|\vec{k}_{\mathrm{eff}}\right| = \left|\vec{k}_1\right| + \left|\vec{k}_2\right|\right)$ is the effective wave vector of Raman beams, T is the time interval between two consecutive pulses, $\vec{g}$ is the acceleration of gravity, and α is the chirp rate of Raman frequency scanning for compensating for the gravity-induced Doppler shift. As the sensitivity depends on T^2, it is highly desirable to make the time interval T as long as possible. However, it reduces the repetition rate of the measurement to use a long T, which is not suitable for rapid gravity measurements. Therefore, a suitable T should be chosen depending on the applications.

A schematic diagram of our experiment is displayed in figure 1. The experiment is realized in a vacuum chamber, which is made of titanium. At the beginning, an intense beam of slow atoms is prepared by a 2D-MOT; details can be found in our previous paper [24]. Those atoms are transported to the main chamber through a 5 mm differential tube by a push beam. About 4×10^8 ^{87}Rb atoms are loaded within 200 ms into the 3D-MOT. Then the magnetic field used to trap the atoms is shut off, and the atoms are further cooled down to 5 μK by polarization-gradient cooling. After adiabatically switching off the lasers, the atoms fall freely in the gravitational field. A bias field of about 180 mG, whose direction is parallel to the Raman beams, is applied for defining a quantization axis. The atoms are selected in a narrow vertical velocity distribution

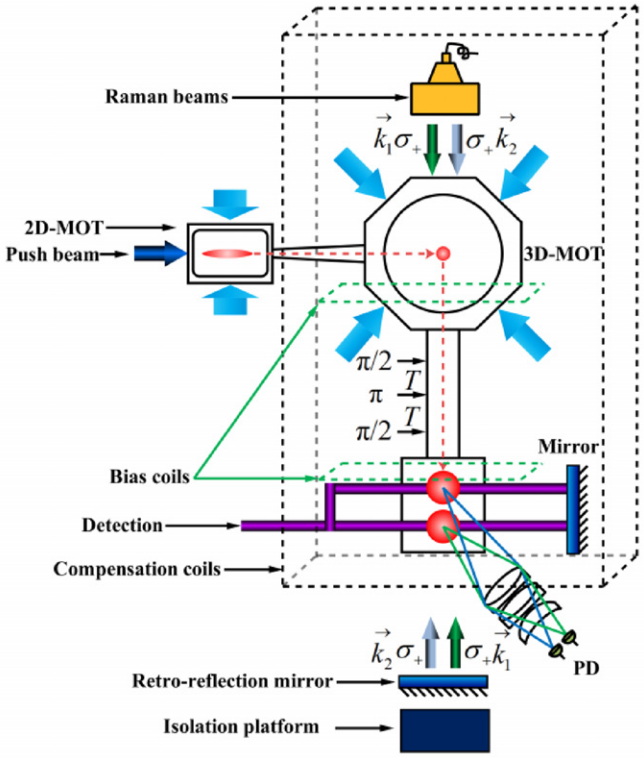


Figure 1. A schematic diagram of our experiment.

in the $|F = 1, m_F = 0\rangle$ state by using a combination of microwave and optical Raman pulses. The durations of the microwave and Raman π pulses for state selection are 1200 μs and 10 μs, respectively. The interferometer is realized with a sequence of three Raman pulses ($\pi/2$–π –$\pi/2$). The total interrogation time $2T$ is 120 ms, and the duration of $\pi/2$ is 5 μs. Finally, the normalized detection system is used to collect the fluorescence at the detection region [25]. The phase of the interferometer can be extracted from the recorded population signal (interference fringe). It takes about 450 ms to complete a single cycle, including atom loading, state preparation, interference procedure and state detection.

A scheme of the Raman laser system is shown in figure 2. The Raman light sources consist of two extended cavity diode lasers (Toptica DL pro). One of them (Raman laser 2) is frequency locked by detecting the beat note with a reference laser, whose frequency is locked by the modulation transfer spectroscopy (MTS) method [26]. The linewidth of the reference laser is about 280 kHz in our experiment. The Raman detuning Δ is set at 1.13 GHz from the $|5^2P_{3/2}, F' = 1\rangle$ excited state. The two Raman beams are phase locked using the optical phase lock loop (OPLL) method. They are overlapped and amplified by the same tapered amplifier (TA). A small part of the amplified laser beam is sent onto a fast photodetector (Hamamatsu G4176) to obtain the beat signal. The beat note is mixed down with a 7 GHz reference frequency. The down-converted signal is compared with a 166 MHz reference frequency from a direct digital synthesizer (DDS) by a fast digital phase frequency detector. Both the 7 GHz frequency source and the DDS are referenced to a 10 MHz oscillator. The frequency of the DDS is also scanned to

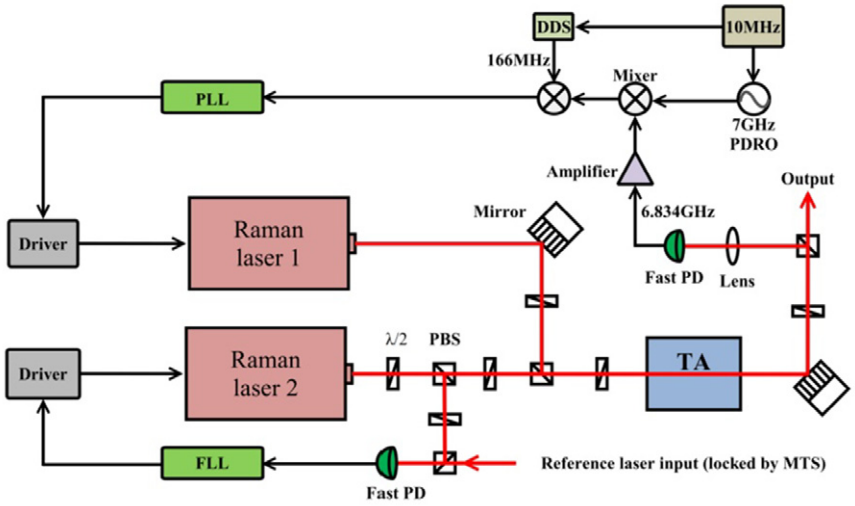


Figure 2. Scheme of Raman laser system. PBS is the polarizing beam splitter, TA is the tapered amplifier, PDRO is the phase-locked dielectric resonator oscillator, PLL is the phase lock loop, FLL is the frequency lock loop and DDS is the direct digital synthesizer.

compensate for the gravity-induced Doppler shift. The OPLL is closed by forming a feedback circuit. A more detailed design and its performance can be found in our previous work [27].

The Raman beams are injected into the same polarization-maintaining fibre and the total power after the fibre is 120 mW. The Raman beams are delivered to the atoms from the top window of the vacuum chamber and retro-reflected by a mirror, which is located on a passive vibration isolation platform with a size of $31 \times 31 \times 11.7\,\mathrm{cm}^3$. This platform is customized by the Minus-K company according to our experimental requirements. It consists of non-magnetic materials and its weight is 16 kg. Its vertical resonant frequency is about 0.5 Hz and the maximal payload is about 14 kg. The contribution of the residual vibration noise of this platform to the interferometer is estimated by weighting the power spectral density of the vibration noise with the transfer function of the interferometer [28], which is about $5.5 \times 10^{-8}\ g$ per shot ($T = 60$ ms).

3. Experimental results

3.1. The atom interference fringe

The gravimeter is operated in the urban environment of Hangzhou city. Our measurement is performed using retro-reflected Raman beams in σ_+–σ_+ polarization configuration. The Doppler-sensitive $\pi/2$–π–$\pi/2$ pulse sequence is implemented. The duration of the Raman π pulse is 10 μs. A homemade DDS is used to produce the frequency chirp of the Raman laser so that the Doppler shift due to gravity can be compensated. The interference fringe of $T = 60$ ms is obtained by varying the chirp rate, which is displayed in figure 3. The phase measurement of the fringe is obtained with a sinusoidal fit of the experimental data. The function used for the fitting is $y = A + B\sin(T^2(x - x_c))$. The fitted uncertainty of the fringe phase is 23 mrad, which corresponds to a resolution of $4 \times 10^{-8}\ g$. The contrast of 31% is mainly limited by the transfer efficiency of the Raman π pulse.

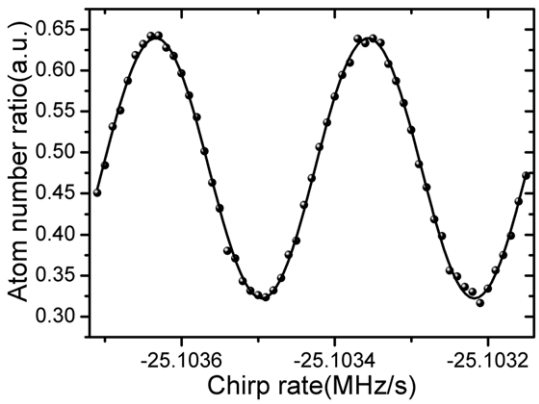


Figure 3. Typical atom interference fringe for $T = 60$ ms. The number of total points is 57 and each point is obtained as an average of four times; one full fringe period corresponds to 102 s.

3.2. The magnetic field system and the evaluation of the quadratic Zeeman phase shift

The magnetic field at the centre of the 3D MOT is well compensated below 1 mG using three pairs of perpendicular Helmholtz coils. These coils yield a cubic profile, which is shown in figure 1. Its outer size is $60 \times 55 \times 80\,\mathrm{cm}^3$ (length × width × height). The coils consist of flat cables with an IDC connector. The flat cables are much lighter and easier to assemble compared with the enamelled wire. Two other rectangular coils, with a size of $40 \times 42\,\mathrm{cm}^2$, are separated by a distance of 40 cm to produce the bias magnetic field. The direction of the bias magnetic field is parallel to the Raman beams, and the amplitude of this magnetic field is controlled by a precise dc power supply. The resonant microwave spectra are measured while varying the falling time of the atoms. The magnitude of the bias magnetic field $B(t)$ is extracted due to the first-order linear Zeeman effect so that the map of the magnetic field in the interaction area could be obtained, which is shown on the left-hand side of figure 4. The phase shift of the interferometer induced by the quadratic Zeeman effect is

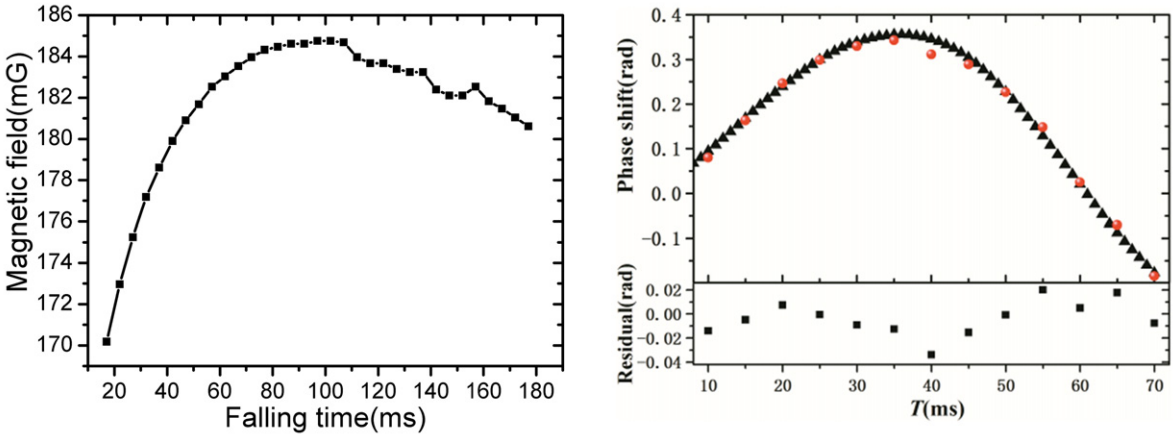


Figure 4. Left: the bias magnetic field measured with a variation of the falling time of the atoms; the applied current of the dc power supply is 0.5 A. Right: the differential phase shift induced by the quadratic Zeeman shift. Black full triangles: the differential shift calculated by the map of the magnetic field. Red full circles: the differential shift measured by inversing the Raman wave vector.

given by

$$\Delta\phi = \int_{-T}^{T} \kappa g_s(t) B^2(t)\, \mathrm{d}t, \tag{2}$$

where κ is the quadratic Zeeman coefficient of the ^{87}Rb ground states D2 line, $g_s(t)$ is the sensitivity function [9]. Two values of the bias magnetic field are selected to extract the differential quadratic Zeeman shift (180 mG and 380 mG). The differential phase shift versus different T calculated by the magnetic field map is shown on the right-hand side of figure 4.

For comparison, we have also measured the magnetic-field-induced phase shifts by the method of inversing the direction of Raman wave vector, which has been described in detail in [29]. Here, we briefly recall this method. When the atoms fall freely after the state selection, the Doppler shift induced by gravity could be compensated for perfectly by the frequency chirp of the Raman beams. There are four Raman beams interacting with the atoms but only two counter-propagating beams could drive the Raman transitions according to the two-photon resonance condition. The Raman transitions with a wave vector pointing upwards ($k_{\mathrm{eff}}\uparrow$) or downwards ($k_{\mathrm{eff}}\downarrow$) depend on the direction of the frequency chirping. So the systematic phase shifts of the interferometer could be separated based on the up or down interferometer configuration, which can be expressed as

$$\Delta\Phi_{k_{\mathrm{eff}}\uparrow} = k_{\mathrm{eff}} g T^2 + \Delta\Phi_{\mathrm{Indep}} + \Delta\Phi_{\mathrm{dep}}, \tag{3}$$

$$\Delta\Phi_{k_{\mathrm{eff}}\downarrow} = -k_{\mathrm{eff}} g T^2 + \Delta\Phi_{\mathrm{Indep}} - \Delta\Phi_{\mathrm{dep}}, \tag{4}$$

where $\Delta\Phi_{\mathrm{Indep}}$ includes the quadratic Zeeman shift, the one-photon light shift and the radiofrequency phase shift, and it could be extracted by

$$\Delta\Phi_{\mathrm{Indep}} = (\Delta\Phi_{k_{\mathrm{eff}}\uparrow} + \Delta\Phi_{k_{\mathrm{eff}}\downarrow})/2. \tag{5}$$

Differential measurements are also carried out to obtain the pure quadratic Zeeman shift. The two values of the applied bias magnetic fields interleave rapidly to yield two different amplitudes of the quadratic Zeeman shift (180 mG and 380 mG), while the other systematic phase shifts are almost the same. Then the difference in $\Delta\Phi_{\mathrm{Indep}}$ of the two magnetic field values gives the pure quadratic Zeeman shift. The differential phase shift versus different values of T is also shown on the right-hand side of figure 4. The measured results coincide well with that calculated by the magnetic field map.

The amplitude of the residuals between the measurements and the calculation is about $\pm$20 mrad, which is shown at the bottom of figure 4. When T is equal to 60 ms, the corresponding measurement uncertainty of the gravitational acceleration is about $\pm$34 μGal. The difference between the measurements and the calculation mainly comes from three factors. Firstly, the experimental data (red full circles) have a measurement uncertainty. As is mentioned in section 3.1, the phase uncertainty of the interference fringe is 23 mrad. Secondly, a perfect rejection of the quadratic Zeeman shift requires a perfect overlap between the atomic trajectories of the up and down interferometer configurations. We have calculated the systematic bias due to the different interferometer configurations with the path integral formalism [30] and it is about 10 mrad in our experiment (T = 60 ms), which could be reduced by applying the symmetry interferometer configurations [31]. Thirdly, the interpolation calculation (black full triangles) will yield an uncertainty due to the fluctuation of the measured magnetic field. The applied magnetic field of 180 mG is measured for more than 12 h to evaluate its stability. The standard deviation of the statistical distribution is 0.031 mG. Taking the uncertainty of each measured magnetic field into account, the uncertainty of the quadratic Zeeman phase shift is estimated to be about 0.1 mrad (T = 60 ms).

Here, it should be noted that the bias magnetic field of 180 mG is not a special value; it mainly depends on the amplitude and the direction of the residual magnetic field in the interference region. Due to the absence of the magnetic shield, this bias magnetic field in our experiment may be larger than the others. In our experiment, if the bias field is smaller, the quantization axis will not be fully defined so that the contrast of the interference fringe will be decreased.

The influence on the accuracy of the gravity measurement without using the magnetic shield could be discussed as

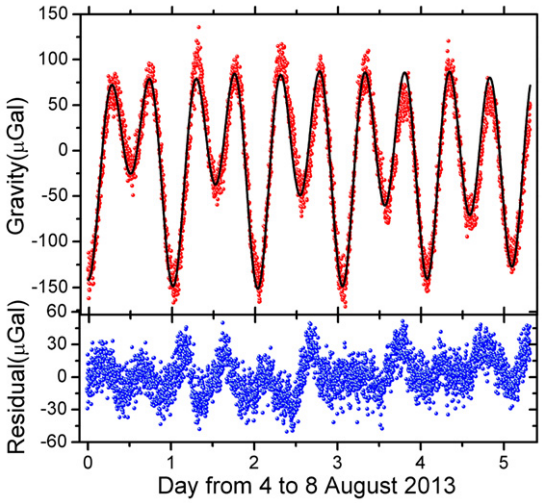


Figure 5. Top: tide data measured from 4 to 8 August 2013 (128 h). Red points represent the experimental data, black solid line is the theoretical tide model without correcting the ocean loading of the Earth. Bottom: residuals (tide data subtracted from the theoretical value).

follows. Firstly, the residual magnetic field in the interference region will be larger if there is no magnetic shield. Therefore, the amplitude of the bias magnetic field should be larger in order to define the quantization axis, which means the offset of the systematic phase shift due to the quadratic Zeeman effect may increase. However, the accuracy of the interpolation calculation could be sufficiently high if the value of the bias field could be measured precisely with enough experimental points and if it is sufficiently stable. Secondly, the influence of the stray magnetic field may increase without the magnetic shield. The accuracy of the gravity measurement may be affected by the stray magnetic field.

3.3. The tidal measurement and the Allan deviation of the gravity measurement

A practical gravimeter should be able to carry out long-term gravity monitoring. Figure 5 shows a continuous gravity measurement of 128 h using our cold atomic gravimeter. The measurement was performed from 4 to 8 August 2013. The complicated change in gravity with time is mainly caused by the tidal force. Each experimental point corresponds to one interference fringe. Each fringe represents 228 drops. The theoretical curve is calculated using the standard tide model, which coincides well with the experimental data. The residuals display the discrepancy between them, which can be seen at the bottom of figure 5. This discrepancy on short time scales is mainly due to the variations of the human activities near our laboratory, which is located in the urban environment of Hangzhou city. The vibration noise is still dominant so that the phase uncertainty of the single interference fringe is about 40 μGal in our experiment. In addition, the measurement has not been corrected for atmospheric pressure, polar motion effect and ocean loading effect [32], which may be another reason for this discrepancy, but this factor is not dominant.

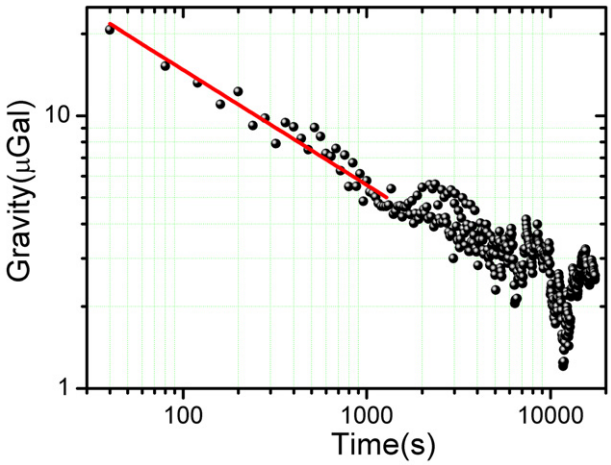


Figure 6. Allan standard deviation of the residuals.

The Allan standard deviation of the residual signal is represented in figure 6. The equivalent sensitivity at 1 s is $1.0 \times 10^{-7}\ g$. A resolution of $1.1 \times 10^{-8}\ g$ is obtained after an integration time of 200 s, and $5.7 \times 10^{-9}\ g$ is achieved within 1000 s integration time.

3.4. The seismic wave measurement

The abnormality of the Ramsey fringe, which was caused by the earthquake of magnitude 7.2 in Pakistan on 28 September 2013, is displayed in figure 7. The duration of this earthquake exceeds 1 h and the whole seismic wave is recorded by our cold gravimeter. It is of great interest to compare our data with the seismic data provided by the Seismological Bureau of the Zhejiang province of China, whose seismic station is less than 1 km away from our laboratory. The acceleration of vibration needs to be extracted from the signal of Ramsey fringes to make this comparison. The acceleration is extracted as follows. Firstly, the experimental data of the interference fringe are fitted by a function of $P(\alpha) = A + B\ \sin((\vec{k}_{\rm eff}\cdot\vec{a} - \alpha)T^2)$. Then the slope κ_i at a given chirp rate α_i is obtained by the derivation of the fitted function ($\kappa_i = {\rm d}P(\alpha)/{\rm d}\alpha$). After subtracting each experimental datum of the interference fringe from the fitted curve, the deviation of the population ΔP could be extracted. The value of $\Delta\alpha$ is given by $\Delta P/\kappa$, and the fluctuation of acceleration reads $\Delta a = \Delta\alpha/k_{\rm eff}$. This fluctuation as a function of time is shown in figure 8.

It is found that our results are in good agreement with the data provided by the Seismological Bureau. The whole set of seismic waves, including shear waves (S waves), compressional waves (P waves) and surface waves, is clearly shown in figure 8. Moreover, the frequencies of the seismic waves are analysed. They have a low frequency of less than 0.1 Hz, and the frequency of the P waves is much higher than that of the S waves and surface waves.

4. Conclusions

In conclusion, we have demonstrated a compact and high-sensitivity atom gravimeter without the heavy magnetic

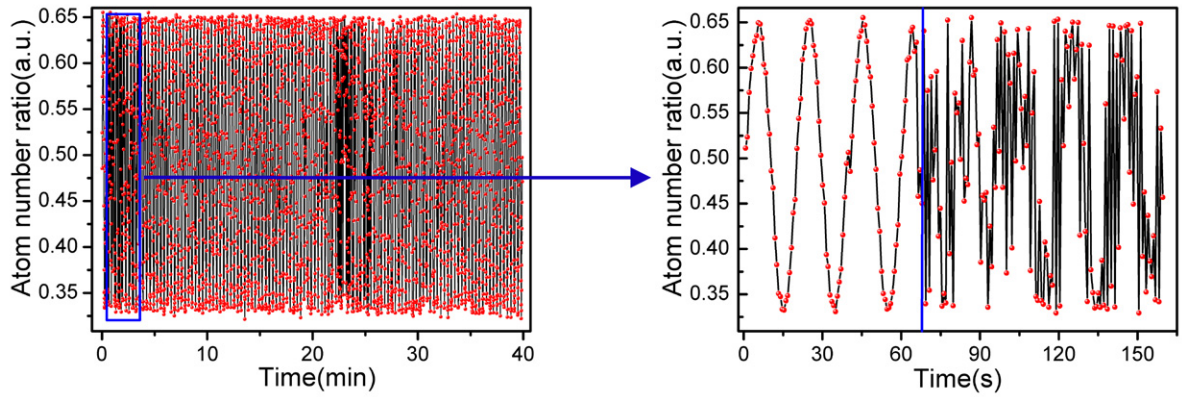


Figure 7. Abnormality of Ramsey fringes during the earthquake of magnitude 7.2 that occurred in Pakistan on 28 September 2013.

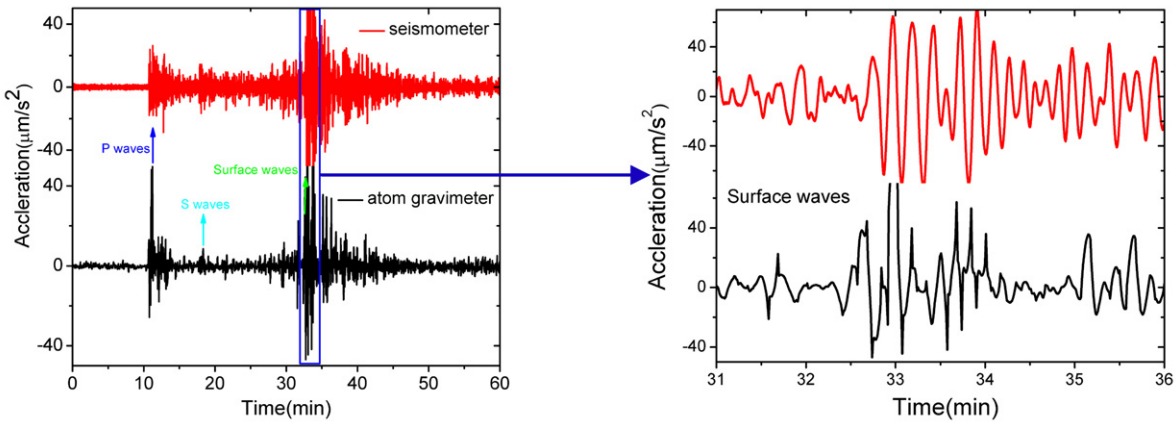


Figure 8. Comparison between our atomic gravimeter and the seismometer nearby. Black curve: the acceleration measured by our atom gravimeter; red curve: the seismic wave signal measured by the traditional seismometer.

shielding system and the active vibration isolation system. The quadratic Zeeman shift was precisely evaluated and discussed in great detail. Despite a short total interrogation time of 120 ms, a sensitivity of $1.0 \times 10^{-7}\ g\,\mathrm{Hz}^{-1/2}$ and a resolution of $5.7 \times 10^{-9}\ g$ within 1000 s integration time were achieved. The long-term gravity data of 128 h were measured and they are in good agreement with the standard tidal model. In addition, a seismic wave of about 1 h has been recorded by our atomic gravimeter. Our atom gravimeter is promising for field applications in the future.

Acknowledgments

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